Cross-regime super-resolution of two-dimensional turbulence: what transfers, what must be chosen per regime, and how to choose it blind

Progress report, 12 August 2026. A diffusion model trained at Reynolds number 1000 reconstructs 256-by-256 turbulent vorticity fields from four-times-coarser observations. This report presents the measured cross-regime behaviour of that model and its reinforcement-learning fine-tunes across Reynolds numbers 1000 to 10000, and the ground-truth-free procedure now in place for choosing the inference configuration at regimes where no reference data exists. All error bars are bootstrapped over independent flow sequences.

1. The experiment behind the figures

Three fine-tuned models (trained at Reynolds numbers 1000, 2000 and 10000 against spectral targets extrapolated from coarse data) and the untuned base model were each pointed at seven regimes, from 1000 to 10000, with the sampling configuration held fixed - so the curves isolate what the model itself carries across regimes. A second pass repeated the exercise with the sampling depth chosen blindly per regime. Grading used 640 held-out triplets per generated regime (16 sequences), or the 70-triplet ceiling of the direct-simulation files. Every cell of an earlier published grading was reproduced digit for digit by this harness before any new claims were made.

2. Spectrum matching across regimes

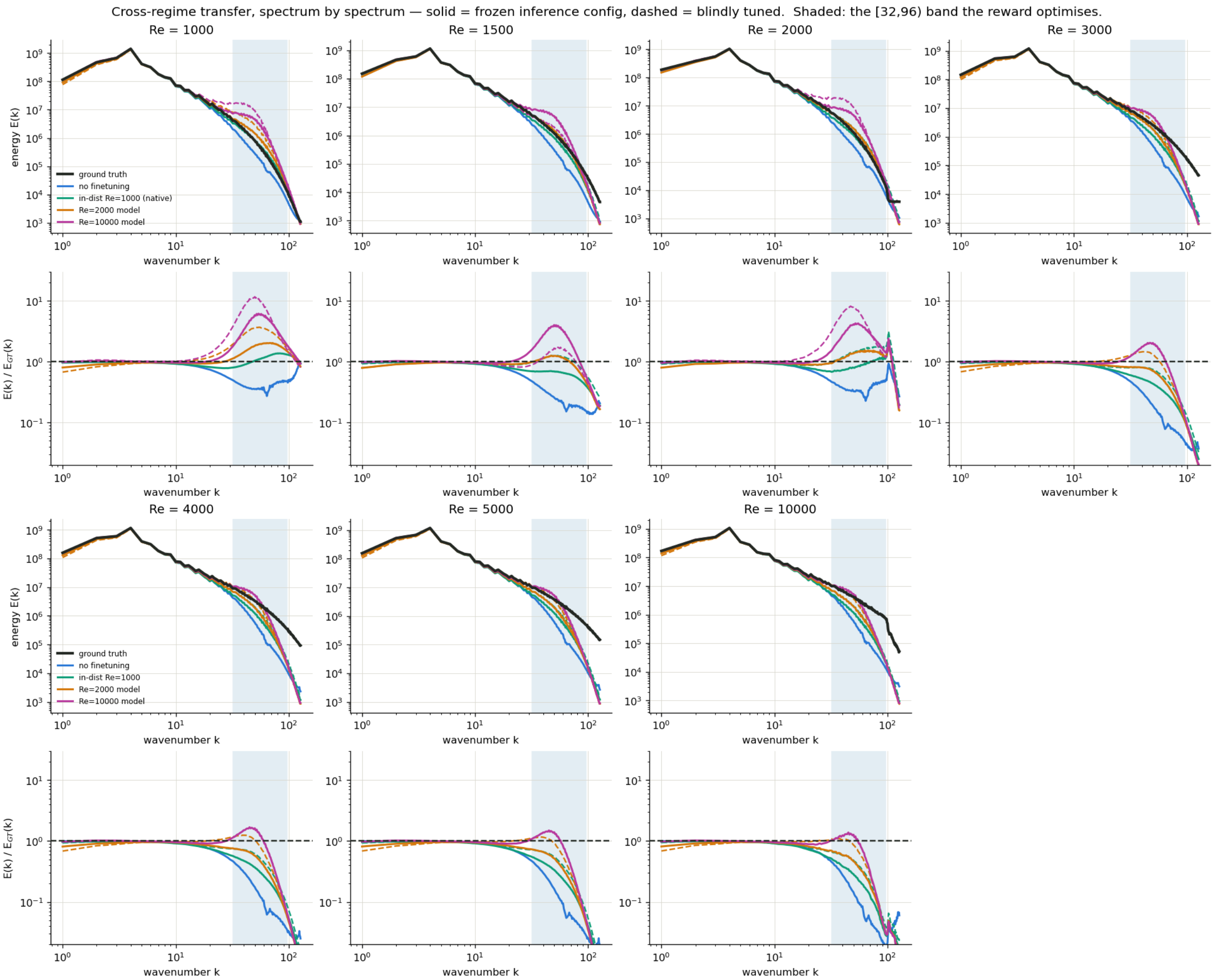

Figure 1. Energy spectra (top of each pair) and their ratio to ground truth (bottom), for every model at every regime. Solid lines: fixed sampling configuration; dashed: blindly tuned depth. Shaded band: wavenumbers 32 to 96, over which retention is scored.

One shape repeats everywhere: models fill the lower half of the scored band and fall away above it - a bump near wavenumber 40 to 60, then a cliff. At Reynolds 1000 the ratio curves sit near one across the whole range; that is what success looks like. By Reynolds 3000 every model is below one tenth of the true energy before wavenumber 128, and the hotter the target, the earlier the cliff. Choosing the depth blindly (dashed) rescales the bump toward one but does not extend the cliff: depth is a gain knob, not a resolution knob.

3. Retention band by band, and the energy-versus-structure trade

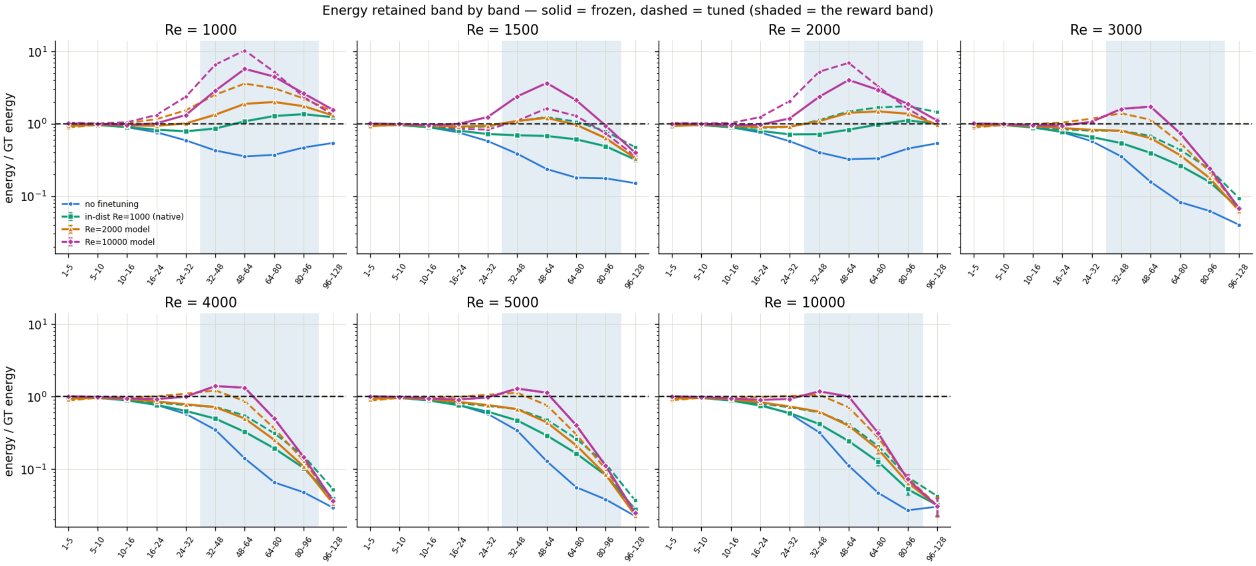

Figure 2. Energy retained relative to truth in ten wavenumber bands. Solid: fixed configuration; dashed: blindly tuned. Shaded: the scored band.

The headline retention numbers hide where the energy sits. The clearest case: the Reynolds 10000 model at Reynolds 3000 reads retention 1.49 - apparently 49 percent too much energy - yet carries only a quarter of the true energy at wavenumbers 80 to 96 and seven percent above 96. It over-fills the bottom of the scored band while the top of the spectrum collapses. Tuning the depth moves the middle bands toward one and leaves the tail almost unchanged (for the Reynolds 2000 model at Reynolds 3000, retention doubles from 0.69 to 1.20 while the band above 80 moves from 0.18 to only 0.22).

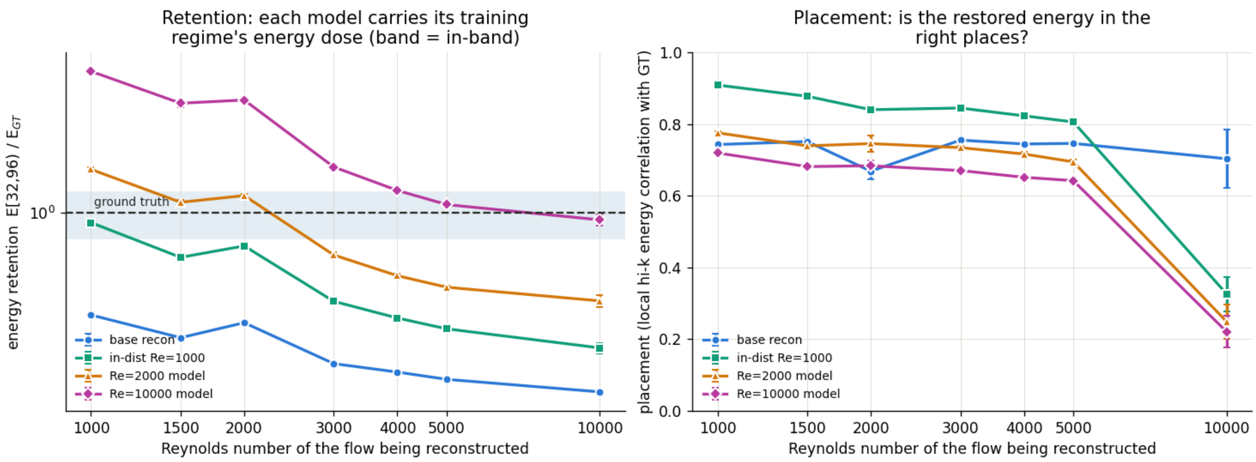

Figure 3. Retention (left) and structural placement (right) against the regime being reconstructed. Placement is the correlation between where the reconstruction puts its fine-scale energy and where the true flow puts it.

Placement falls monotonically as a model's energy dose rises, in every regime, in both directions of adaptation. The only model that ever places better than the untuned base is an under-dosed one. Above roughly ten times the training Reynolds number placement collapses outright (0.22 against the base's 0.70) even where retention looks correct - and the collapse is not caused by the amount restored: at Reynolds 10000 even a model that restores almost nothing loses half its placement. Below the ceiling, the first increment of restored energy lands in the right places; above it, even the first increment is misplaced.

4. The physics residual, read correctly

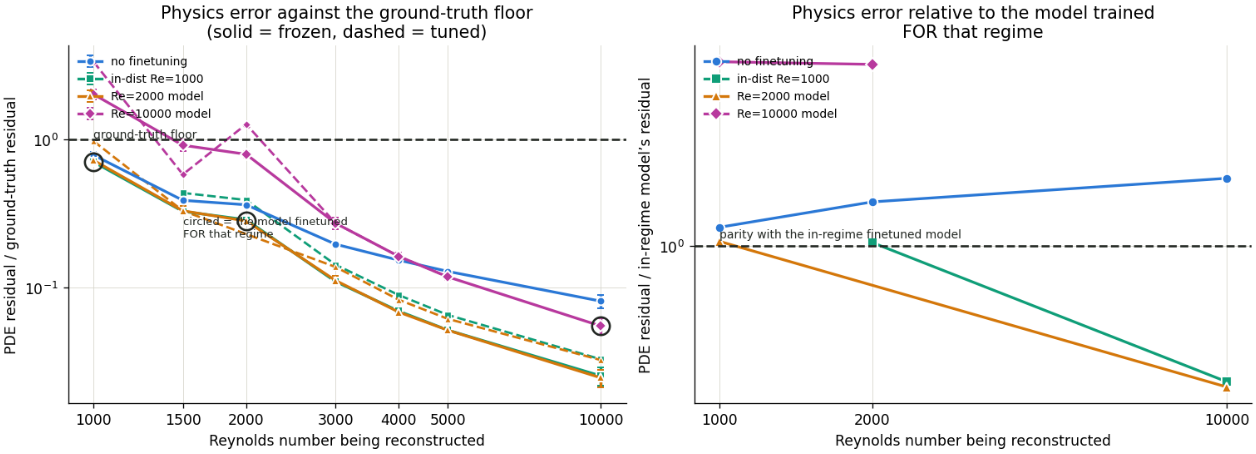

Figure 4. Left: Navier-Stokes residual of each reconstruction divided by the ground truth's own residual, which rises steeply with Reynolds number (16 at Re 1000 to 878 at Re 10000). Circles mark each regime's native model. Right: the same residual relative to the model fine-tuned for that regime.

Reconstructions sit below the truth's own residual almost everywhere - smooth fields have small derivatives - so a low residual is not fidelity. The residual is a one-sided instrument: among under-restored models it is nearly blind (models forty percent apart in retention read identically at three separate regimes), and it only speaks once a model reaches or exceeds the correct dose. Two further cautions are recorded: the frozen residual-versus-Reynolds law used in the reward is wrong (its exponent should be roughly 1.8, not 2.7 - a saturation effect of the finite grid), and the residual is sensitive to the evaluation batch shape, so only matched-batch comparisons are quoted.

5. What the reconstructions look like

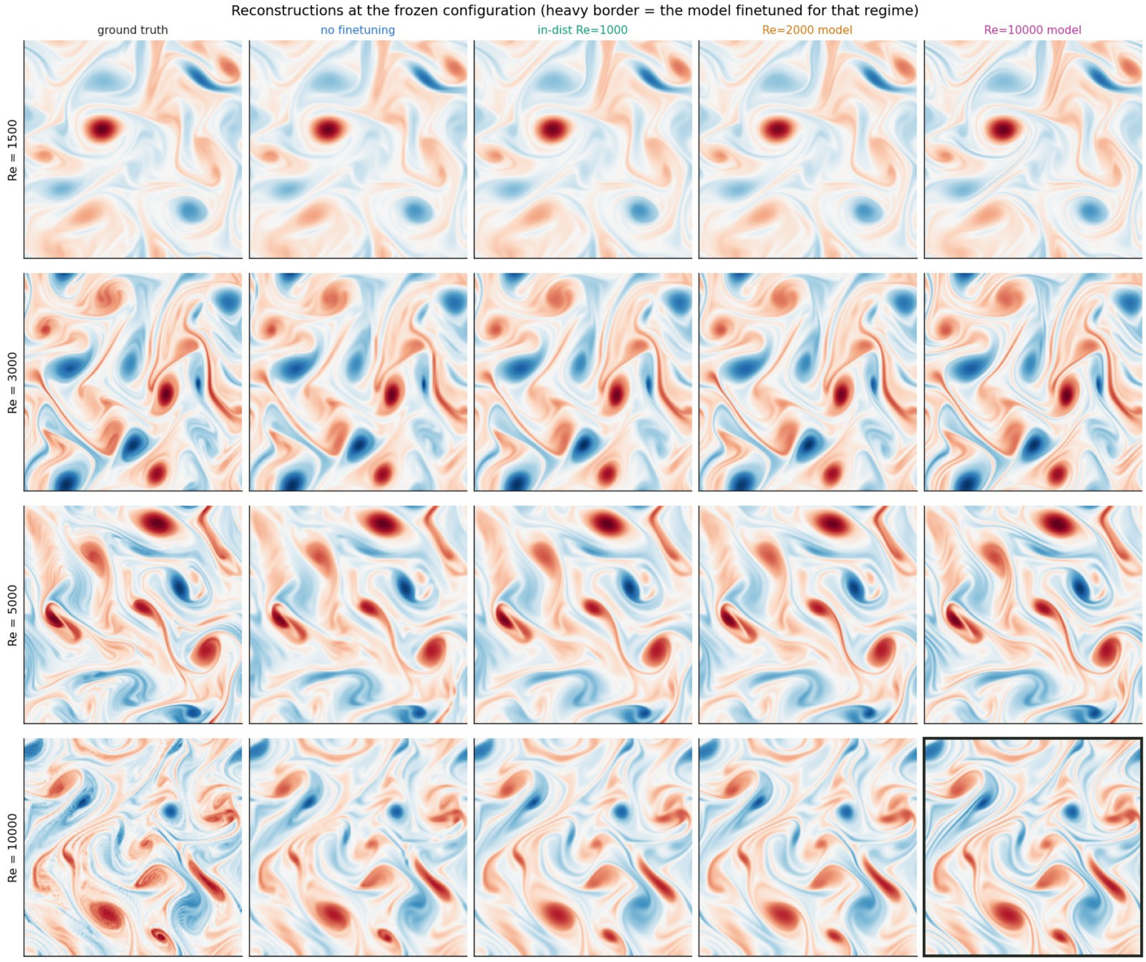

Figure 5. Vorticity fields at four regimes: ground truth, no fine-tuning, and the three fine-tuned models, at the fixed configuration. Heavy border: the model native to that regime.

6. Do the fine-tuned models transfer? The direct answer

Every fine-tuned model was pointed at four regimes none of them had seen, with the configuration frozen - transfer with no retuning at all. Entries are retention / placement (640 held-out triplets per regime). Earlier-generation models throughout this table.

Unseen regime	No fine-tune	Re 1000 model	Re 2000 model	Re 10000 model (old)
1500	0.34 / 0.75	0.68 / 0.88	1.10 / 0.74	2.60 / 0.68
3000	0.27 / 0.76	0.46 / 0.84	0.69 / 0.73	1.49 / 0.67
4000	0.25 / 0.74	0.40 / 0.82	0.58 / 0.72	1.22 / 0.65
5000	0.24 / 0.75	0.37 / 0.81	0.52 / 0.69	1.08 / 0.64

Read along any row: each model carries its training regime's energy dose in its weights. Pointed below that dose it over-delivers (the old Reynolds 10000 model gives two and a half times the required energy at 1500); above it, it under-delivers. Which model is best at a regime is decided purely by dose distance - so raw transfer works only between neighbouring regimes, and fails predictably otherwise.

Transfer becomes genuinely useful when the sampling depth is chosen per regime from coarse data alone. The same models, blindly depth-adapted:

Case (earlier-generation models)	frozen	blindly adapted
Re 2000 model across 1500 / 3000 / 4000 / 5000	1.10 / 0.69 / 0.58 / 0.52	1.10 / 1.20 / 0.98 / 0.88 - all within the acceptance bar
Old Re 10000 model brought DOWN to 1500	2.60 (2.6x over)	1.21, placement 0.68 to 0.78
Re 1000 model pushed UP beyond 3000	0.46-0.37	no depth suffices - correctly reported as out of reach befo

Eleven of twelve blind adaptation decisions were correct or improving. One fine-tune plus a coarse-data depth search covered a three-point-three-fold Reynolds range that would otherwise have needed four separate fine-tunes. The current selection experiment (section 9) extends this: the Reynolds 1000 fine-tune, borrowed at every research regime, delivers roughly four times the untuned base's retention everywhere tested so far.

7. The earlier Reynolds 10000 campaign, kept on the record

Clearly labelled as earlier-generation work: the model was trained on the first-generation file whose training pool contained only eight usable triplets, and that file was later shown spectrally deficient at high wavenumber. With that stated: the sealed one-shot test - configuration frozen before unsealing, test sequences opened once - PASSED, retention 0.758 (0.792 on the clean subset) inside the pre-registered bar, effective resolution raised from 32 to 61, large scales intact at 0.99. At the maximum available sample (35 sequences) retention reads 0.934. The structural finding stands regardless of data vintage: placement collapsed to 0.22 against the untuned base's 0.70 - the information ceiling near ten times the training Reynolds number - and the later decomposition showed the collapse occurs even for models that restore almost nothing, so it is a property of the regime distance, not of over-commitment. The regime is being redone on new high-fidelity 512-squared data (one long sequence in hand, more in generation) with clean train, validation and test splits.

8. Which knob matters: the cascade-depth-by-checkpoint grids

At the two regimes whose ground truth we own (Reynolds 500 and 1000), every combination of nine sampling depths and thirteen model states was graded - 126 cells in total. Three results now anchor the deployment procedure.

Finding	Measurement
Depth dominates the checkpoint	retention spread 0.30 across depths vs 0.08 across checkpoints (Re 500); 1.02 vs 0.45 (Re 1
Regimes want opposite depths	Re 500 is healthy only at the five-chain cascade (retention 1.01); Re 1000 overshoots there c
Fine-tuning is not always worth it	at Re 500 the untuned base at the right depth beats every fine-tuned checkpoint on retention

A further methodological result: the total restored energy alone cannot define a healthy model. Twelve cells at Reynolds 500 sit within ten percent of the perfect energy total while their structural quality spans from excellent to unusable (one has the correct band total and an effective resolution of one). Health is therefore defined by energy and placement jointly wherever truth allows both to be measured.

9. Choosing the configuration without ground truth

The deployable procedure, frozen before use and documented in full elsewhere: build the spectral anchor from the target's coarse data by the fixed recipe; score every candidate (model state times depth) by the ratio of its output energy to the anchor's over wavenumbers 10 to 95; shortlist the cells reading 0.77 to 0.85 - the interval in which verified-healthy models read at both calibration regimes (0.784 and 0.835); discard candidates whose fine-scale energy map has drifted far from the base reconstruction's (a truth-free structural guard, validated at correlation 0.96 and 0.74 against true placement); take the survivor nearest 0.81, breaking ties toward shallower depth and less training; finally splice the base reconstruction's large scales below a truth-free crossover, which restores them to within one percent. The untuned base is always a candidate, so 'do not fine-tune' is a permitted outcome.

10. The ablation, completed: was the second calibration regime worth it? No.

Two copies of the selection procedure were run at eight research regimes (Reynolds 1500 to 8000), identical except that one calibrated its target interval from Reynolds 1000 alone and the other from both calibration regimes. Candidates included the Reynolds 1000 fine-tune's two best checkpoints at every regime, and at Reynolds 2000 twelve natively trained checkpoints as well. Selections were made blind; afterwards each pick was graded against held-out data. The final verdict, all eight regimes:

Regime	Best untuned (oracle)	Single-regime arm picked	Both-regimes arm picked
1500	0.397	K3, borrowed 449: ret 0.75, place 0.83	K3, borrowed 149: ret 0.72, place 0.87
2000	0.329	K3, NATIVE ckpt 149: ret 0.82, place 0.84	same pick
3000	0.257	K4, borrowed 149: ret 0.72, place 0.79	K5, borrowed 449: ret 1.13, place 0.66
4000	0.224	K5, borrowed 449: ret 0.92, place 0.65	same pick
5000	0.205	K5, borrowed 449: ret 0.82, place 0.64	same pick
6000	0.197	K5, borrowed 149: ret 0.74, place 0.69	same pick
7000	0.188	K5, borrowed 149: ret 0.67, place 0.67	same pick
8000	0.182	K5, borrowed 149: ret 0.62, place 0.66	same pick

The two arms made identical decisions at seven of the eight regimes, including the decisive Reynolds 2000 grid where fifteen model states competed and both arms passed over the borrowed models for the native early checkpoint (graded retention 0.817, placement 0.84, effective resolution 84 - the same depth-and-early-checkpoint shape as the verified healthy Reynolds 1000 cell). The single divergence went against the better-informed arm: at Reynolds 3000 its higher target pushed one depth rung deeper into a thirteen-percent spectral overshoot with damaged fine-scale structure. Conclusion: calibrating at a second ground-truth regime bought confirmation, not better decisions - the selection rule does not need more ground-truth regimes to work. Two further readings: the borrowed Reynolds 1000 fine-tune delivers two to four and a half times the best the untuned base achieves at any depth, with retention decaying smoothly from 0.92 at Reynolds 4000 to 0.62 at Reynolds 8000 - a visible reach limit rather than a cliff. And one caveat: at every regime except Reynolds 2000 the structural guard vetoed all in-band candidates (its reference drifts with regime), so those picks are nearest-to-band fallbacks; they graded well regardless, but the guard needs re-specifying before the rule is declared final.

11. Data provenance and limits

All nine research regimes were regenerated through a single verified pipeline (1024-squared direct simulation, spectrally reduced to 256-squared; per-sequence independent initial conditions; frame spacing matching the training convention), replacing earlier mixed-provenance files whose generator differences had corrupted two extrapolation laws. Known limits stand: the anchor runs progressively hot with Reynolds number (up to roughly nineteen percent), the calibration interval rests on two regimes at or below the training Reynolds number, and structural placement remains unmeasurable without truth above the information ceiling near ten times the training regime. New high-fidelity Reynolds 10000 data (one long 512-squared sequence, more being generated) will re-open that regime with clean train, validation and test splits.